

Follow-Up Investigation: 80m EFHW 2:14 Winding Ratio

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Project: 80m End-Fed Half-Wave (EFHW) Antenna
Investigation Focus: Revised winding ration from 3:21 to 2:14
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This document serves as a follow-up investigation to the previous EFHW antenna evaluation and documents the quantitative results after rewinding the 49:1 UNUN transformer to the recommended 2:14 turn ratio configuration. The primary objective of this iteration was to validate whether the revised winding ratio would improve transformer behavior and produce more consistent resonant characteristics across the HF spectrum. Only the winding ratio was modified.

Transformer Revision

The updated transformer configuration consists of:

- Ferrite Core: Type 31 toroid
- Transformer Ratio: 2 primary turns : 14 secondary turns
- Transformer Type: 49:1 UNUN
- Capacitor: 100pF across the primary winding
- Antenna Configuration: 80m EFHW deployment

The prior investigation indicated potential instability associated with alternate winding configurations and the effects of the ferrite type 31, including a 3:21 arrangement. The revised winding follows the more conventional 49:1 EFHW transformer topology.

Physical Configuration

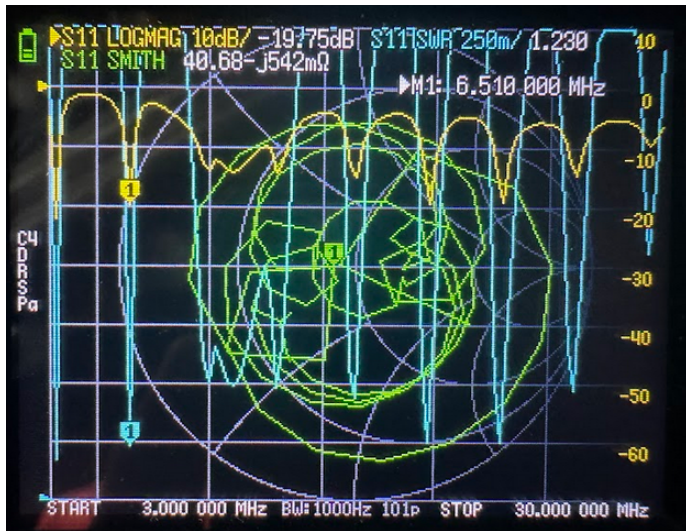


Shows the revised 49:1 UNUN wound with the updated 2:14 turn ratio.

Wideband Sweep Results (3–30 MHz)

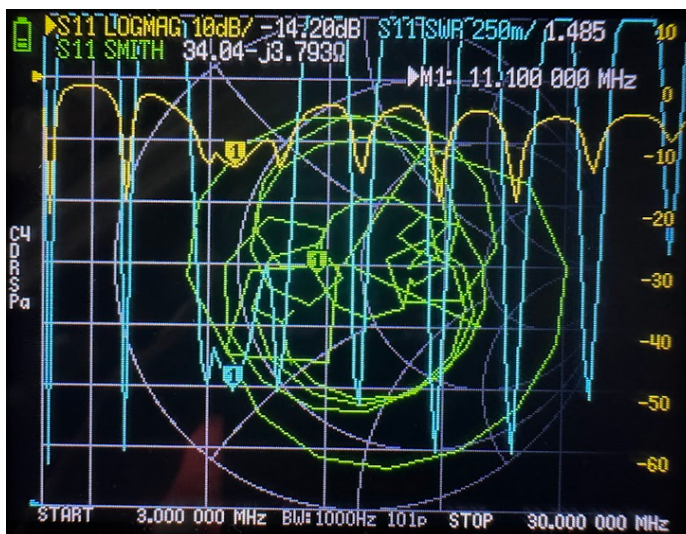
The following measurements were captured using the NanoVNA across the 3–30 MHz spectrum.
Resonant Dips Observed

Picture	Frequency	SWR
	3.27 MHz	1.173



6.51 MHz

1.230



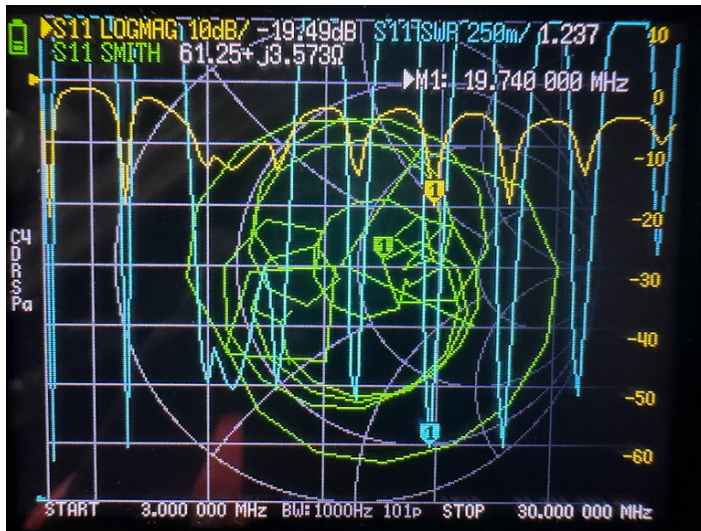
11.10 MHz

1.485



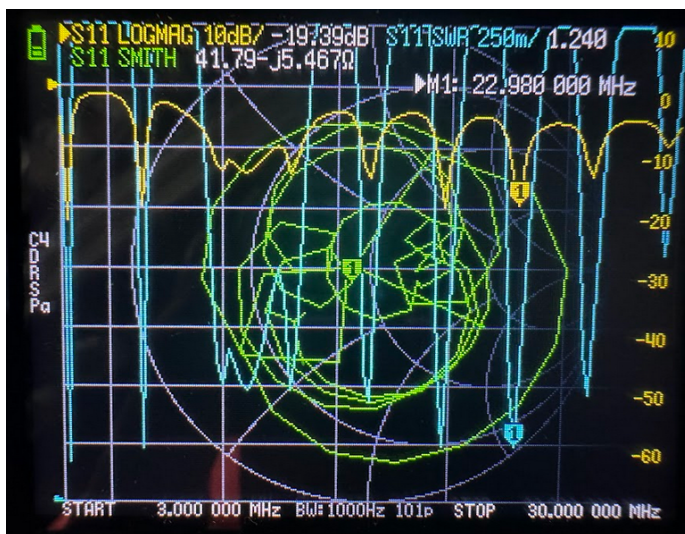
16.50 MHz

1.436



19.74 MHz

1.237



22.09 MHz

1.240



26.22 MHz

1.462

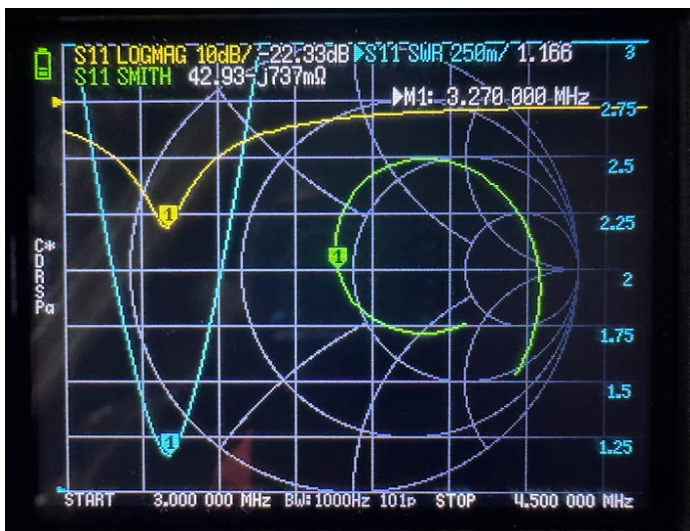


The resulting sweep demonstrates a significantly cleaner and more predictable harmonic structure than previous measurements obtained using the alternate winding arrangement.

The primary resonance at 3.27 MHz confirms successful operation within the 80 meter region. Additional harmonic resonances appear throughout the HF spectrum, though several do not align precisely with amateur allocations.

Band-Specific Investigation

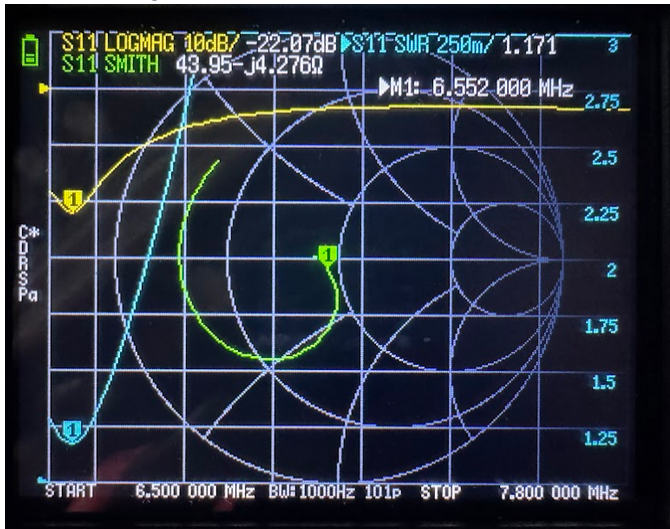
80 Meter Band



- Sweep Range: 3–4.5 MHz
- Minimum SWR: 1.166
- Resonant Frequency: 3.27 MHz

This result demonstrates excellent matching performance within the 80 meter range. The resonance remains slightly below the traditional 80 meter phone segment, suggesting the antenna is electrically long and may require trimming to shift the resonance upward.

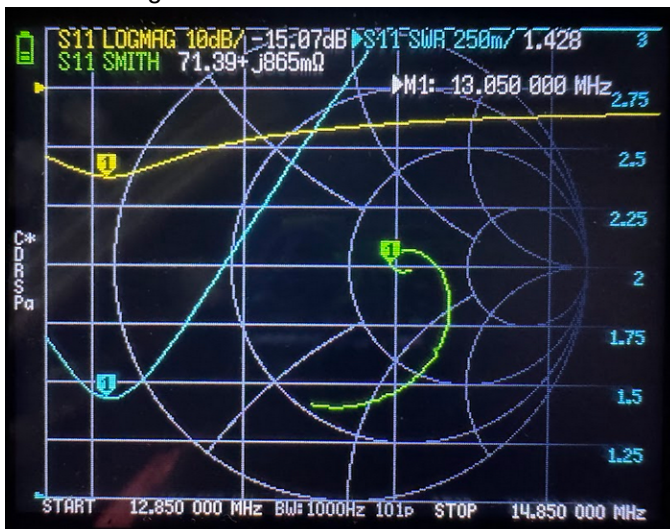
40 Meter Region



- Sweep Range: 6.5–7.8 MHz
- Minimum SWR: 1.171
- Resonant Frequency: 6.552 MHz

A secondary harmonic resonance appears near the 40 meter allocation. This resonance demonstrates low SWR and confirms the expected harmonic behavior of the EFHW structure.

20 Meter Region



- Sweep Range: 12.85–14.85 MHz
- Minimum SWR: 1.428
- Resonant Frequency: 13.05 MHz

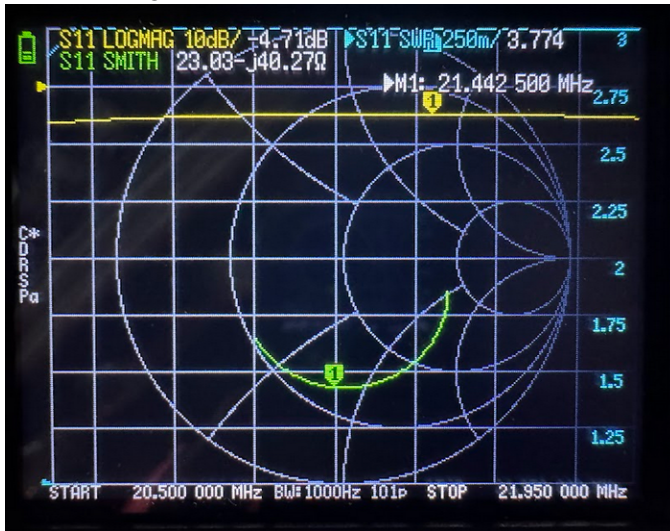
The observed resonance does not align with the amateur 20 meter allocation. Instead, it appears substantially lower in frequency than expected.

This indicates the harmonic spacing is compressed relative to an ideal EFHW harmonic structure. Potential contributing factors include:

- Wire length
- Loading effects
- Transformer parasitics
- Counterpoise interaction
- Deployment geometry
- Ground coupling

Despite the misalignment, the SWR remains relatively acceptable.

15 Meter Region



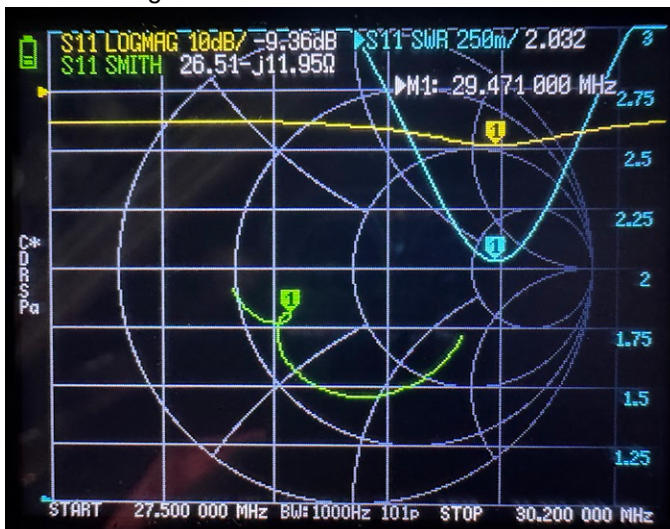
- Sweep Range: 20.5–21.95 MHz
- Minimum SWR: 3.774
- Resonant Frequency: 21.442 MHz

The 15 meter region demonstrates degraded matching performance. Although resonance exists within the band, the SWR rises significantly compared to the lower harmonics.

This may indicate:

- Increased transformer losses at higher harmonics
- Common-mode current effects
- Harmonic distortion caused by deployment geometry
- Insufficient choking or counterpoise optimization

10 Meter Region



- Sweep Range: 27.7–30.2 MHz
- Minimum SWR: 2.032
- Resonant Frequency: 29.471 MHz

The 10 meter resonance falls near the upper end of the band and demonstrates moderate mismatch.

Compared to the previous transformer configuration, however, the resonance structure appears considerably more stable and repeatable.

Comparative Observations

Compared to the earlier investigation using the 3:21 winding ratio:

- Harmonic resonances appear more stable
- The transformer behavior is more predictable
- The SWR curve is smoother
- The 80 meter match improved substantially
- High-frequency behavior remains imperfect but improved

- The overall system now resembles a more conventional EFHW response

The revised winding ratio appears to have corrected several impedance transformation issues observed previously.

Plausible Interpretation of Why?

The earlier 3:21 configuration likely introduced excessive inductance and distributed capacitance into the transformer, especially when combined with the lossy characteristics of the Type 31 ferrite core. Since Type 31 is highly effective as a broadband choking material, the additional turns may have caused the transformer to partially behave like an RF choke, suppressing or “choking out” higher-frequency energy rather than efficiently transferring it.

The revised 2:14 configuration reduces winding length and parasitic effects, allowing the transformer to operate closer to the intended 49:1 impedance transformation with cleaner harmonic behavior and improved resonance stability.

Next Steps for Improvement

1. Install with type 43 ferrite material.
Although results are significantly better, type 43 is still preferred.
2. Shift the 80m Resonance Upward
The primary resonance at 3.27 MHz is slightly low for the intended 80 meter operating region.
 - Incrementally shorten the antenna wire
 - Retest after each adjustment
 - Target desired operating segment
3. Evaluate Deployment Geometry
The current sloper arrangement introduces environmental asymmetry.
 - Compare sloper versus inverted-L configurations
 - Raise feedpoint height
 - Reduce near-ground wire interaction
 - Evaluate endpoint height effects

Conclusion

The revised 2:14 transformer winding configuration represents a substantial improvement over the previous implementation.

The antenna now demonstrates:

- Stronger 80 meter matching
- Cleaner harmonic structure
- More stable resonant behavior
- Improved transformer performance consistency